Level 1: Fundamentals (The "Must-Knows")

What is the difference between a Docker Image and a Container?

Answer: An image is a read-only blueprint (class), while a container is a running instance (object) of that image.

How does Docker differ from a Virtual Machine (VM)?

Answer: VMs virtualize hardware and include a full Guest OS. Docker virtualizes the OS kernel, sharing the host OS, making containers much lighter and faster.

What are the main components of the Docker Architecture?

Answer: Docker Client, Docker Host (Daemon), and Docker Registry.

Explain the Docker container lifecycle.

Answer: Created → Running → Paused/Unpaused → Stopped → Deleted.

What is a Dockerfile, and what are its most common instructions?

Answer: A script containing commands to assemble an image. Key instructions: FROM, RUN, COPY, EXPOSE, CMD, and ENTRYPOINT.

What is the difference between CMD and ENTRYPOINT?

Answer: ENTRYPOINT sets the main executable that cannot be easily overridden, while CMD provides default arguments that are easily replaced by the user at runtime.

How do you see the logs of a crashed container?

Answer: Use docker logs <container_id>. If it's not running, add the -a flag to docker ps to find the ID first.

What is Docker Hub?

Answer: A public cloud-based registry for sharing and storing Docker images.

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Level 2: Storage, Networking & Compose

What is the difference between a Volume and a Bind Mount?

Answer: Volumes are managed by Docker (stored in /var/lib/docker/volumes). Bind mounts link to any specific path on the host machine.

Explain the default Docker network types.

Answer: Bridge (default for single hosts), Host (removes isolation between host and container), and None (no networking).

What is Docker Compose and why do we use it?

Answer: A tool for defining and running multi-container applications using a single YAML file.

How can containers communicate with each other?

Answer: By placing them on the same user-defined bridge network; they can then reach each other using their container names as hostnames.

What is a "Dangling Image"?

Answer: An image that is no longer associated with a tagged image (shows as <none>:<none>). You clean them with docker image prune.

How do you map a port from the host to a container?

Answer: Using the -p flag, e.g., docker run -p 8080:80 nginx (Host port 8080 maps to Container port 80).

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Level 3: Advanced & Scenario-Based

What is a Multi-stage build and why is it useful?

Answer: It uses multiple FROM statements in one Dockerfile to compile code in one stage and copy only the binary to a smaller final stage, drastically reducing image size.

How do you handle sensitive data (passwords) in Docker?

Answer: Use Docker Secrets (in Swarm mode) or Environment Variables/Environment Files, though Secrets are more secure.

What are "Namespaces" and "Cgroups" in Docker?

Answer: Namespaces provide isolation (Process, Network, Mount). Cgroups (Control Groups) manage resource limits (CPU, Memory).

A container is running but you can't access the service. How do you troubleshoot?

Answer: Check docker ps (is it running?), docker logs (app errors?), docker port (port mapping?), and docker exec (is the service listening inside?).

What is Docker Swarm and how does it differ from Kubernetes?

Answer: Swarm is Docker's native orchestration tool (easier to setup). Kubernetes is more complex but more powerful for large-scale enterprise automation.

How do you limit a container's memory and CPU usage?

Answer: Use flags during docker run: --memory="512m" and --cpus="1.5".

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